

Exam Date & Time: 09-Jan-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - JANUARY-2024

Number Theory and Cryptography [CSE-3111]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compute all congruence classes of Modulo 8
Illustrate extended Euclidian algorithm for finding the inverse of 12 mod 79 (5)
 - A)
 - B) a. Calculate $\phi(n)$ for $n = 1200$ and $n = 2008$.
b. Let $n \in \mathbb{N}$ and let p be a prime. Show that if $p|n$ then $\phi(n \cdot p) = p \cdot \phi(n)$. (5)
- 2) Prove that $n^{13} - n$ is divisible by 2, 3, 5, 7, and 13 for any integer n . (5)
 - A)
 - B) Let ' m ' be an odd integer and let ' a ' be any integer. Prove that $2m + 2a$ can never be a perfect square. (5)
- 3) Find the remainder when $2^{100} + 3^{100} + 4^{100} + \dots + 10^{100}$ is divide by 11. (5)
 - A)
 - B) Let X be a nonempty set. Show that the power set $R = 2^X$ together with the set operations of symmetric difference $A + B = (A \cap \bar{B}) \cup (B \cap \bar{A})$ and intersection $AB = A \cap B$, is a ring. Is R commutative? Does it have unity? Justify (5)
- 4) Encrypt the message "meet me at the usual place at ten rather than eight o clock" using the Hill cipher with the key= $\begin{pmatrix} 7 & 3 \\ 2 & 5 \end{pmatrix}$ (5)
 - A) a. Show your calculations and the result.
b. Show the calculations for the corresponding decryption of the ciphertext to recover the original plaintext. (5)
 - B) Using the Vigenere cipher, encrypt the word "cryptographic" using the word "mit". Analyse the efficiency of mono & poly alphabetic cipher (5)
- 5) Consider the system which uses highly simplified DES Algorithm with Iteration-4 output as "1A037825BD81C028" (hex). Compute Iteration-5 using the key as "13902B148936" (hex). (5)

A)

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07
1	00	15	07	04	14	02	13	10	03	06	12	11	09	05	03	08
2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00
3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13

B)

Justify, why S-Box plays a vital role in DES. State the encryption process of double and triple DES. Discuss with an example.

(5)

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